Sharp pointwise bounds for Lebesgue functions

Qiyuan Gu
University of Chicago
phoenix1203@uchicago.edu

Abstract

For each n, let λ_n be the Lebesgue function for polynomial interpolation at an arbitrary set of n distinct nodes in $[-1,1]$. We prove that the points $x \in (-1,1)$ for which $\lambda_n(x) > (2/\pi)\log n - C(x)$ infinitely often, with some finite $C(x)$, form a dense set. We also prove $\limsup_{n\to\infty} \lambda_n(x)/\log n \geq 2/\pi$ for almost every $x \in (-1,1)$. We also show that no absolute additive constant works for all arrays: for every $M > 0$, there is an array for which each fixed $x \in (-1,1)$ eventually satisfies $\lambda_n(x) \leq (2/\pi)\log n - M$. For the same array, no single finite constant gives a dense set of points with infinitely many lower-bound occurrences. The positive results use derivative-jump energies and positive Cauchy transforms. The counterexample combines an explicit logarithmic integral estimate on the gaps of a compact set with real-rooted polynomials having prescribed amplitudes.

1 Introduction

For each $n \geq 1$, let

$$-1 \leq x_{1,n} < \cdots < x_{n,n} \leq 1.$$

Define the node polynomial, the cardinal polynomials, and the Lebesgue function by

$$P_n(x) = \prod_{k=1}^{n} (x - x_{k,n}),$$

$$\ell_{k,n}(x) = \frac{P_n(x)}{(x - x_{k,n})P_n'(x_{k,n})}, \qquad \lambda_n(x) = \sum_{k=1}^{n} |\ell_{k,n}(x)|.$$

The cardinal polynomials are interpreted by continuity at the nodes. The rows of the array need not be nested.

Theorem 1. *For every triangular array of distinct nodes as above, the following statements hold.*

(i) The set of points $x \in (-1,1)$ for which there is a finite constant $C(x)$ such that

$$\lambda_n(x) > \frac{2}{\pi}\log n - C(x) \tag{1.1}$$

for infinitely many n is dense in $(-1,1)$.

(ii) For Lebesgue almost every $x \in (-1,1)$,

$$\limsup_{n\to\infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}. \tag{1.2}$$

Erdős posed the existence and almost-everywhere questions for triangular arrays in [4, p. 68], and they are recorded as Problem 1132 in [2]. The constant in (1.1) is independent of n; it may depend on the array and the point x. As Tao notes in [9, Remark 1.12], the original question leaves the allowed dependence of the additive constant unspecified. The following result separates two uniform interpretations from Theorem 1(i).

Theorem 2. *For every $M > 0$, there is a triangular array of distinct nodes in $(-1, 1)$ such that, for every fixed $x \in (-1, 1)$,*

$$\lambda_n(x) \leq \frac{2}{\pi} \log n - M \qquad \textit{for all sufficiently large } n. \tag{1.3}$$

The starting index may depend on x. The array can also be chosen so that, for every finite C, the set

$$G_C = \left\{ x \in (-1, 1) : \lambda_n(x) > \frac{2}{\pi} \log n - C \textit{ for infinitely many } n \right\} \tag{1.4}$$

is not dense in $(-1, 1)$.

Thus no absolute C guarantees even one point with infinitely many lower-bound occurrences for every array. Even if C is allowed to depend on the array, it need not give a dense good-point set. These statements concern arbitrary triangular arrays, as in [4, p. 68] and [9, Corollary 1.11]. The construction does not impose nesting between rows, so it does not give a counterexample for the initial segments of a single infinite sequence.

Faber [8] proved the order $\log n$ for the global Lebesgue constant, Bernstein [1] obtained the asymptotically sharp leading constant $2/\pi$, Erdős and Turán [6] improved the error to $O(\log \log n)$, Erdős [5] to $O(1)$, and Vértesi [10, Corollary 1] refined the asymptotics through the optimal constant term to an error of $O((\log \log n / \log n)^2)$. See [9, Section 1.3] and [3] for further historical discussion.

For triangular arrays, Erdős [4, p. 68] stated that the set of points x satisfying

$$\limsup_{n \to \infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}$$

is everywhere dense, and asked whether this holds for almost every x. Erdős and Vértesi [7] proved that for every triangular array there is a continuous function whose interpolants diverge almost everywhere; in particular, $\limsup_{n \to \infty} \lambda_n(x) = \infty$ for almost every x. Thus Theorem 1(ii) gives the sharp logarithmic rate almost everywhere.

Tao [9, Theorem 1.10(i)] proved that every fixed nondegenerate interval $I \subset [-1, 1]$ satisfies

$$\sup_{x \in I} \lambda_n(x) \geq \frac{2}{\pi} \log n - O_I(1).$$

His Corollary 1.11 and Remark 1.12 give, for every prescribed $\omega(n) \to \infty$, a dense, indeed comeager, set of points x for which

$$\lambda_n(x) \geq \frac{2}{\pi} \log n - \omega(n)$$

infinitely often. Theorem 1(i) replaces $\omega(n)$ by a finite constant depending on the point, on a dense set. Baire category enters (i) at one point, to produce an interval with a uniform upper bound; recurrence on a positive-measure set follows from the energy estimate in Section 3.

Theorem 2 is proved in Section 7, independently of the preceding lower-bound arguments. A compact set with many small gaps supports an explicit continuous function whose logarithmic integral ratio is large in every gap. Smooth positive approximations also make this ratio large at each fixed point of the compact set. An amplitude construction converts these functions into interpolation arrays.

Throughout the paper, all logarithms are natural and $|E|$ denotes Lebesgue measure. We write $c_0 = 2/\pi$, and suppress the row index when discussing a single row. Constants in local estimates may depend on the fixed intervals and test functions, but not on n or on interpolation data bounded by one. The notation $J \Subset U$ means that J is a compact subset of the open set U.

2 A local differentiation formula

We use the following local form of Riesz's differentiation formula and its near-equality consequence.

Lemma 3. *Suppose that F is holomorphic in a neighborhood of the square $\{|\Re z|, |\Im z| \le T\}$, is real on its real diameter, and satisfies, throughout this square,*

$$|F(u+iv)| \le e^{|v|}.$$

Then

$$|F'(0)| \le 1 + O(T^{-1}). \tag{2.1}$$

If $F'(0) \ge 1-\delta$, where $\delta \ge 0$, then

$$F(\pi/2) \ge 1 - \frac{\pi^2}{4}\delta - O(T^{-1}). \tag{2.2}$$

The constants are absolute.

Proof. Choose $R = m\pi$ with $T/2 \le R < T$, and integrate

$$\frac{F(z)}{z^2 \cos z}$$

around the square of radius R. On its vertical sides, $|\cos z| = \cosh(\Im z)$; on its horizontal sides, $|\cos z| \ge \sinh R$. The contour integral, divided by $2\pi i$, is consequently $O(R^{-1})$. Taking residues at zero and at $t_j = (j + \frac{1}{2})\pi$ gives

$$F'(0) = \sum_{|t_j|<R} \frac{(-1)^j F(t_j)}{t_j^2} + O(T^{-1}). \tag{2.3}$$

Since $|F(t_j)| \le 1$ and

$$\sum_{j\in\mathbb{Z}} \frac{1}{t_j^2} = 1,$$

equation (2.1) follows. Isolating the term $t_0 = \pi/2$ in (2.3) gives

$$1 - \delta \le \frac{4}{\pi^2} F(\pi/2) + 1 - \frac{4}{\pi^2} + O(T^{-1}),$$

which proves (2.2). □

We apply this formula to interpolants with bounded nodal data. The estimates we use from Tao are stated first, with their normalization:

$$S_n = \sum_k \frac{1}{|P_n'(x_{k,n})|}, \qquad \alpha_n = \frac{1}{n}\log S_n, \qquad U_n(z) = \frac{1}{n}\log\frac{1}{|P_n(z)|}.$$

Proposition (Tao's local estimates). *Let $I \Subset (-1,1)$ be a fixed nondegenerate closed interval, and suppose that $\sup_I \lambda_n \ll n^{O(1)}$. With α_n normalized as above [9, equation (4.11)], there are densities ρ_n on $\operatorname{int} I$ with the following properties. For each fixed compact interval $I' \Subset \operatorname{int} I$:*

1. *The densities are bounded above and below by positive constants and have uniformly bounded Lipschitz norms:*

$$\rho_n(x) \asymp_{I'} 1, \qquad |\rho_n(x) - \rho_n(x_0)| \ll_{I'} |x - x_0| \qquad (x, x_0 \in I').$$

   This is [9, Theorem 4.1(ii), equation (4.6)].

2. *For $x \in I'$ and $\eta \geq \log n/n$,*

$$U_n(x + i\eta) = \alpha_n - \pi\eta\rho_n(x) + O_{I'}\left(\frac{\log n}{n} + \eta^3\right). \tag{2.4}$$

   This is [9, equation (4.7)].

3. *For every subinterval $K \subset I'$ and every $A \geq 1$,*

$$\frac{1}{n}\#\{k : x_{k,n} \in K\} = \int_K \rho_n(x)\,dx + O_{I'}\left(\frac{|K|}{A} + \frac{A^2 \log n}{n}\right). \tag{2.5}$$

   This is [9, Theorem 4.1(iv), equation (4.9)].

The function ρ_n here is Tao's density defined from the original interval I by his equation (4.15), so it does not depend on the auxiliary choice of I'. Taking $A = n^{1/10}$ in (2.5) and integrating by parts shows that, for functions h_n supported in a fixed compact subset of $\operatorname{int} I$ with uniformly bounded Lipschitz norms,

$$\frac{1}{n}\sum_k h_n(x_{k,n}) = \int h_n(x)\rho_n(x)\,dx + O(n^{-1/10}). \tag{2.6}$$

Lemma 4. *Let $I \Subset (-1,1)$ be a fixed nondegenerate closed interval. Suppose that, for all sufficiently large n,*

$$\lambda_n(x) \leq \Lambda_n := c_0 \log n + C \qquad (x \in I), \tag{2.7}$$

where C is fixed. This bound gives the hypothesis of the proposition above; let ρ_n be its densities. For every fixed interval $J \Subset \operatorname{int} I$ and every real interpolant

$$q(x) = \sum_{k=1}^{n} a_k \ell_{k,n}(x), \qquad |a_k| \leq 1,$$

uniformly for $x \in J$,

$$|q'(x)| \leq \pi n \rho_n(x)(\Lambda_n + o(1)), \qquad |q''(x)| \leq C_J n^2 \log n. \tag{2.8}$$

Moreover, for every fixed D, if

$$\frac{|q'(x)|}{\pi n \rho_n(x)} \geq c_0 \log n - D, \tag{2.9}$$

then there is $y = x + O_J(n^{-1})$ such that

$$\lambda_n(y) \geq c_0 \log n - D', \tag{2.10}$$

where D' depends on I, J, C, D, but not on n, x, q. The point y lies in any fixed compact interior interval containing J in its interior, for all sufficiently large n.

Proof. Fix $I' \Subset \operatorname{int} I$ with $J \Subset \operatorname{int} I'$. For $\Re z \in I'$ and $\Im z = \eta \geq \log n/n$, the interpolation formula and (2.4) give

$$|q(z)| \leq \frac{S_n|P_n(z)|}{\eta} \leq \frac{1}{\eta} \exp\Big(\pi n\eta\rho_n(\Re z) + O(\log n + n\eta^3)\Big). \tag{2.11}$$

Fix $x \in J$, and put

$$L = n^{-1/5}, \qquad H = n^{-2/5}, \qquad \omega = \pi n\rho_n(x)(1 + n^{-1/10}).$$

On the upper edge of $\{|\Re z - x| \leq L,\ 0 \leq \Im z \leq H\}$, we have $|q(z)| \leq \Lambda_n e^{\omega H}$: ωH exceeds $\pi n H\rho_n(x)$ by $\pi\rho_n(x)n^{1/2}$, which dominates the $O(n^{2/5})$ from the Lipschitz variation of ρ_n and the remaining $O(\log n)$ terms in (2.11). On the lower edge $|q| \leq \Lambda_n$; a Chebyshev expansion on I gives $|q| \leq \Lambda_n \exp(O_I(n))$ on the vertical sides. Apply [9, Lemma 2.5] to $\Lambda_n^{-1}q(z)e^{i\omega(z-x)}$. At distance at least $L/2$ from the vertical sides, its error factor is $\exp(O(ne^{-cL/H})) = 1 + O(n^{-10})$. Thus, also reflecting across the real axis,

$$|q(x + u + iv)| \leq \Lambda_n(1 + n^{-10})e^{\omega|v|} \qquad (|u| \leq L/2,\ |v| \leq H). \tag{2.12}$$

The rescaled function

$$F(t) = \frac{q(x + t/\omega)}{\Lambda_n(1 + n^{-10})}$$

satisfies Lemma 3 on a square of radius $T \asymp_J n^{3/5}$. By (2.1),

$$|q'(x)| = \omega\Lambda_n(1 + n^{-10})|F'(0)| \leq \pi n\rho_n(x)\Lambda_n(1 + n^{-1/10})(1 + O(n^{-3/5})).$$

Since $\Lambda_n n^{-1/10} = o(1)$, this gives the first bound in (2.8); Cauchy's estimate on a disk of radius comparable to n^{-1} gives the second. Under (2.9), change the sign of q if necessary. For sufficiently large n,

$$F'(0) \geq \frac{c_0 \log n - D}{\Lambda_n(1 + n^{-1/10})(1 + n^{-10})} \geq 1 - \frac{C_1}{\Lambda_n}, \qquad C_1 = 1 + \max\{C + D, 0\}.$$

Equation (2.2) then gives $|q(x + \pi/(2\omega))| \geq \Lambda_n - C_2$. Since $|q(y)| \leq \lambda_n(y)$ on the real axis and $\omega \asymp_J n$, this proves (2.10) with the asserted displacement. $\square$

3 Derivative jumps and a Riesz energy

At each internal interpolation node, write

$$J_k = \lambda_n'(x_{k,n}+) - \lambda_n'(x_{k,n}-).$$

At the first and last nodes, use the exterior polynomial pieces. Set

$$\nu_k = \frac{1}{S_n|P_n'(x_{k,n})|}.$$

The absolute values of all cardinal polynomials other than $\ell_{k,n}$ have a cusp at $x_{k,n}$, whereas $\ell_{k,n}$ is positive near that node. Using $\ell_{j,n}(x) = P_n(x)/((x - x_{j,n})P_n'(x_{j,n}))$ and $P_n(x_{k,n}) = 0$ to differentiate at $x_{k,n}$, we obtain

$$J_k = 2\sum_{j \neq k} |\ell_{j,n}'(x_{k,n})| = 2\sum_{j \neq k} \frac{\nu_j}{\nu_k|x_{k,n} - x_{j,n}|}. \tag{3.1}$$

Lemma 5. *Under the hypotheses of Lemma 4, choose a nonzero nonnegative smooth function f compactly supported in* $\operatorname{int} I$*. Then*

$$\sum_k \frac{f(x_{k,n})^2}{n\rho_n(x_{k,n})} \frac{J_k}{2\pi n\rho_n(x_{k,n})} \geq c_0 \log n \int f(x)^2\,dx - O(1). \tag{3.2}$$

Terms with $f(x_{k,n}) = 0$ are taken to be zero; ρ_n is only needed on a fixed interior neighborhood of the support of f.

Proof. Put $a_k = f(x_{k,n})/\rho_n(x_{k,n})$, with $a_k = 0$ outside the support of f, and define

$$\mu_n = \frac{1}{n}\sum_k a_k \delta_{x_{k,n}}, \qquad K_\varepsilon(t) = \frac{1}{\sqrt{t^2+\varepsilon^2}}.$$

The kernel K_ε is positive definite, since

$$K_\varepsilon(t) = \frac{1}{\sqrt{\pi}}\int_0^\infty s^{-1/2} e^{-s\varepsilon^2} e^{-st^2}\,ds$$

is a positive mixture of Gaussian kernels. Write

$$\mathcal{E}_\varepsilon(\sigma,\tau) = \iint K_\varepsilon(x-y)\,d\sigma(x)\,d\tau(y)$$

for its bilinear energy. Positivity applied to $\mu_n - f(x)\,dx$ gives

$$\mathcal{E}_\varepsilon(\mu_n,\mu_n) \geq 2\mathcal{E}_\varepsilon(\mu_n, f\,dx) - \mathcal{E}_\varepsilon(f\,dx, f\,dx). \tag{3.3}$$

Take $\varepsilon = 1/n$. Subtracting $f(x)$ in the integral over $|t| \leq 1$, and using the Lipschitz bound for f, gives the uniform estimate

$$(K_{1/n} * f)(x) = 2\log n\, f(x) + O_f(1). \tag{3.4}$$

The mass of μ_n is bounded. Also, by (2.6),

$$\int f\,d\mu_n = \frac{1}{n}\sum_k \frac{f(x_{k,n})^2}{\rho_n(x_{k,n})} = \int f^2 + O(n^{-1/10}). \tag{3.5}$$

Again by (3.4),

$$\mathcal{E}_{1/n}(f\,dx, f\,dx) = 2\log n \int f^2 + O(1).$$

Together with (3.3)–(3.5), this yields

$$\mathcal{E}_{1/n}(\mu_n,\mu_n) \geq 2\log n \int f^2 - O(1). \tag{3.6}$$

The diagonal part of this energy is $n^{-1}\sum_k a_k^2 = O(1)$. Since $K_{1/n}(t) \leq 1/|t|$ for $t \neq 0$,

$$\frac{1}{n^2}\sum_{k\neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq 2\log n \int f^2 - O(1). \tag{3.7}$$

Finally,

$$a_k^2 \frac{\nu_j}{\nu_k} + a_j^2 \frac{\nu_k}{\nu_j} \geq 2a_k a_j.$$

By (3.1), the left side of (3.2) is

$$\frac{1}{\pi n^2}\sum_{k\neq j}\frac{a_k^2\nu_j}{\nu_k|x_{k,n}-x_{j,n}|}.$$

Pairing the terms indexed by (k,j) and (j,k), and then applying (3.7), bounds this expression below by

$$\frac{1}{\pi n^2}\sum_{k\neq j}\frac{a_k a_j}{|x_{k,n}-x_{j,n}|}\geq\frac{2}{\pi}\log n\int f^2-O(1).$$

This proves (3.2). □

4 A dense set of points with bounded additive loss

Proposition 6. *Assume the local upper bound* (2.7) *on a fixed nondegenerate interval* $I\Subset(-1,1)$*. There are a constant* D *and a measurable set* $G\subset\operatorname{int}I$*, with* $|G|>0$*, such that every* $x\in G$ *satisfies*

$$\lambda_n(x)\geq c_0\log n-D$$

for infinitely many n*.*

Proof. Choose compact intervals $J_0\Subset J_1\Subset\operatorname{int}I$, and choose f as in Lemma 5 with $\operatorname{supp}f\subset J_0$. All constants below are independent of n: after I,C,J_0,J_1,f are fixed, we choose D, then $\kappa,D_1,d,\kappa_1,d_1$, and finally b,D_2. Let

$$w_k=\frac{f(x_{k,n})^2}{n\rho_n(x_{k,n})},\qquad T_k=\frac{J_k}{2\pi n\rho_n(x_{k,n})},\qquad W=\int f^2>0.$$

By (2.6),

$$\sum_k w_k=W+O(n^{-1/10}),\qquad 0\leq w_k\leq C_f/n. \tag{4.1}$$

On each interval between successive nodes, λ_n equals an interpolant with nodal data in $\{-1,1\}$. Both one-sided derivatives at a node therefore satisfy (2.8), so

$$0\leq T_k\leq\Lambda_n+o(1) \tag{4.2}$$

uniformly on the support of f. Lemma 5, (4.1), and (4.2) show that

$$\sum_k w_k(\Lambda_n+1-T_k)=O(1), \tag{4.3}$$

with nonnegative summands for all sufficiently large n. On a node with $T_k<c_0\log n-D$, the factor in (4.3) is greater than $C+1+D$. Hence, by choosing D sufficiently large, such nodes carry at most $W/2$ of the weights. Since (4.1) has total weight $W+o(1)$, for all sufficiently large n the nodes in the support of f satisfying

$$T_k\geq c_0\log n-D \tag{4.4}$$

carry at least $W/4$ of the weights. In particular, there are at least κn such nodes, with $\kappa>0$ independent of n. Discarding the first and last nodes does not change this conclusion after reducing κ.

At a node satisfying (4.4), at least one adjacent polynomial piece q has

$$|q'(x_{k,n})|\geq\pi n\rho_n(x_{k,n})(c_0\log n-D).$$

Lemma 4 gives a point

$$y_k = x_{k,n} + O(n^{-1}), \qquad \lambda_n(y_k) \geq c_0 \log n - D_1. \tag{4.5}$$

The same polynomial piece q has value 1 at both endpoints of the corresponding gap. Rolle's theorem and (2.8) imply that this gap has length at least d/n for some fixed $d > 0$ when the gap is contained in J_1. If it is not contained in J_1, its length is bounded below by $\operatorname{dist}(J_0, \mathbb{R} \setminus J_1) > 0$.

An interval of length less than d/n contains at most two such nodes, so every interval of length $O(n^{-1})$ contains only a bounded number of the nodes in (4.4). By (4.5), the same holds, with multiplicities, for the points y_k. Hence one can select at least $\kappa_1 n$ of them, separated by d_1/n, for fixed $\kappa_1, d_1 > 0$.

Equation (2.8), applied to every polynomial piece, also shows that λ_n is $C_I n \log n$-Lipschitz on a fixed compact interior interval containing these points. Choose a fixed sufficiently small $b > 0$, and let A_n be the union of the intervals of radius

$$r_n = \frac{b}{n \log n}$$

centered at the selected points. For all sufficiently large n, these intervals are disjoint, lie in a fixed compact interior interval, and satisfy

$$\lambda_n(x) \geq c_0 \log n - D_2 \quad (x \in A_n), \qquad \frac{c}{\log n} \leq |A_n| \leq \frac{C_2}{\log n}. \tag{4.6}$$

The separation of the centers in the m-th row implies, for every interval B,

$$|A_m \cap B| \leq C_3 \left(\frac{|B|}{\log m} + \frac{1}{m \log m} \right).$$

Summing this inequality over the at most n components of A_n gives, for $m > n$,

$$|A_n \cap A_m| \leq C_3 \left(\frac{|A_n|}{\log m} + \frac{n}{m \log m} \right). \tag{4.7}$$

Put $B_j = A_{2^j}$. Equations (4.6) and (4.7) give

$$|B_j| \asymp j^{-1}, \qquad |B_j \cap B_k| \leq C_4 \left(\frac{1}{jk} + \frac{2^{j-k}}{k} \right) \quad (j < k). \tag{4.8}$$

For every fixed j_0, the sum of the measures of B_j, $j_0 \leq j \leq N$, grows at least as a positive constant times $\log N$. By (4.8), the integral of the square of the sum of their indicators is $O((\log N)^2)$. Cauchy–Schwarz consequently gives a positive lower bound, independent of j_0, for

$$\left| \bigcup_{j \geq j_0} B_j \right|.$$

Continuity of measure along these decreasing tail unions shows that

$$|\limsup_{j \to \infty} B_j| > 0. \tag{4.9}$$

Together with (4.6), this proves the proposition. $\square$

Proof of Theorem 1(i). Fix a nondegenerate compact interval $K_0 \subset (-1,1)$, and suppose that it contains no point satisfying the assertion. Then

$$\lambda_n(x) - c_0 \log n \longrightarrow -\infty \qquad \text{for every } x \in K_0. \tag{4.10}$$

The closed sets

$$F_M = \{x \in K_0 : \lambda_n(x) \le c_0 \log n + M \text{ for every } n \ge 2\}, \qquad M = 1, 2, \dots,$$

cover K_0, by (4.10). Baire's theorem supplies a nondegenerate closed subinterval $I \subset K_0$ on which (2.7) holds with a fixed C. Proposition 6 contradicts (4.10). Thus every such K_0 contains a good point, proving density. Increasing the final additive constant by one gives the strict inequality in the statement. □

5 Positive Cauchy transforms

Suppress the row index and put

$$S = \sum_k \frac{1}{|P'(x_k)|}, \qquad \nu_k = \frac{1}{S|P'(x_k)|}, \qquad \epsilon_k = \operatorname{sgn} P'(x_k).$$

Define

$$m(z) = \sum_k \frac{\nu_k}{z - x_k}, \qquad g(z) = \sum_k \frac{\epsilon_k \nu_k}{z - x_k} = \frac{1}{SP(z)}. \tag{5.1}$$

The last identity is the partial fraction formula for $1/P$, whose expansion at infinity gives

$$\sum_k \epsilon_k \nu_k x_k^j = 0 \qquad (0 \le j \le n-2). \tag{5.2}$$

For $n \ge 2$, both $m + g$ and $m - g$ are Cauchy transforms of positive probability measures: their masses at x_k are $(1 \pm \epsilon_k)\nu_k$.

For real x away from the nodes, write

$$H(x) = \sum_k \frac{\nu_k}{|x - x_k|}, \qquad \gamma_s(x) = -\Im m(x + is), \qquad p_s(t) = \frac{s}{\pi(t^2 + s^2)}.$$

Then

$$\lambda_n(x) = \frac{H(x)}{|g(x)|}, \qquad H(x) = \frac{2}{\pi} \int_0^\infty \gamma_s(x)\, \frac{ds}{s}, \tag{5.3}$$

and the Poisson semigroup gives

$$\gamma_s = p_{s-h} * \gamma_h \qquad (s > h). \tag{5.4}$$

Lemma 7. *For every row with $n \ge 2$, every $0 < h \le 1$, and every $x \in [-1,1]$,*

$$\frac{|g(x + ih)|}{\gamma_h(x)} \le \frac{4 + h^2}{h^2 \sinh((n-1)h/2)}. \tag{5.5}$$

Consequently, if $0 < a < 1$ and $h = n^{-a}$, then

$$\sup_{x \in [-1,1]} \frac{|g(x + ih)|}{\gamma_h(x)} \longrightarrow 0. \tag{5.6}$$

Proof. Set $z = x + ih$ and $p = T_{n-1}$, the Chebyshev polynomial of the first kind. The polynomial $(p(t) - p(z))/(t - z)$ has degree at most $n - 2$, so (5.2) gives

$$p(z)g(z) = \sum_k \frac{\epsilon_k \nu_k p(x_k)}{z - x_k}, \qquad |p(z)g(z)| \le \frac{1}{h},$$

using $|p(x_k)| \le 1$ and $\sum_k \nu_k = 1$. Write $z = \cos(u + iv)$, with $u, v \in \mathbb{R}$. Then $h \le \sinh|v|$, so $|v| \ge \operatorname{arsinh} h \ge h/2$. The identity $T_{n-1}(\cos w) = \cos((n-1)w)$ yields

$$|p(z)|^2 = \cos^2((n-1)u) + \sinh^2((n-1)v) \ge \sinh^2((n-1)h/2).$$

Finally, since the nodes and x lie in $[-1, 1]$,

$$\frac{h}{4 + h^2} \le \gamma_h(x) = \sum_k \frac{\nu_k h}{(x - x_k)^2 + h^2}.$$

Combining these bounds proves (5.5). For $h = n^{-a}$, its right-hand side tends to zero: the hyperbolic sine grows exponentially in n^{1-a}, while h^{-2} grows polynomially. This proves (5.6). $\square$

Lemma 8. *Let F be the Cauchy transform of a positive probability measure with finite support. For every bounded real interval S_0, every $x \in \mathbb{R}$, and every $h > 0$,*

$$\int_{\mathbb{R}} p_h(x - y)\, \mathbf{1}_{\{F(y) \in S_0\}}\, dy = \frac{1}{\pi} \int_{S_0} \frac{-\Im F(x + ih)}{(t - \Re F(x + ih))^2 + (\Im F(x + ih))^2}\, dt. \tag{5.7}$$

Proof. The function F maps the upper half-plane into the lower half-plane. Compose harmonic measure of S_0 there with F. The resulting bounded harmonic function on the upper half-plane has the indicated boundary values almost everywhere. The exceptional boundary points are poles of F and preimages of the endpoints of S_0, a finite set. Its Poisson representation is (5.7). $\square$

6 The almost-everywhere bound

Proof of Theorem 1(ii). The failure set is the union, over rational $0 < c < c_0$ and integers $N \ge 2$, of the measurable sets

$$\{y \in (-1, 1) : \lambda_n(y) \le c \log n \text{ for every } n \ge N\}.$$

If the assertion fails, one of these sets E has positive measure. For its fixed c, N, we therefore have

$$\lambda_n(y) \le c \log n \qquad (y \in E,\ n \ge N). \tag{6.1}$$

Choose constants

$$K > 1, \qquad 0 < b < a < 1, \qquad \frac{2b}{\pi c K} > 1,$$

which is possible because $c < 2/\pi$. Set

$$h = n^{-a}, \qquad \eta = n^{-b}, \qquad H^\eta(x) = \frac{2}{\pi} \int_\eta^1 \gamma_s(x)\, \frac{ds}{s}.$$

By (5.3), $H^\eta \le H$. For $s > 0$ and real x, y, t, the triangle inequality gives

$$|y + is - t| \le |x + is - t| + |x - y| \le \left(1 + \frac{|x - y|}{s}\right) |x + is - t|.$$

Taking reciprocal squares and summing against the positive masses yields $\gamma_s(x) \le (1 + |x - y|/s)^2 \gamma_s(y)$. Integrating over $\eta \le s \le 1$, we obtain

$$H^\eta(x) \le \left(1 + \frac{|x-y|}{\eta}\right)^2 H^\eta(y). \tag{6.2}$$

Fix $x \in E$, and write

$$u = \Re m(x + ih), \qquad \gamma = \gamma_h(x), \qquad S_x = [u - K\gamma, u + K\gamma].$$

By (5.6) and (5.7), each of the two events

$$m(y) + g(y) \in S_x, \qquad m(y) - g(y) \in S_x$$

has probability

$$p_K + o(1), \qquad p_K = \frac{2}{\pi} \arctan K > \frac{1}{2},$$

under the Poisson probability $p_h(x-y)\,dy$. The error is uniform in $x \in E$: for $F = m \pm g$, (5.6) gives $\Re F(x + ih) = u + o(\gamma)$ and $-\Im F(x+ih) = \gamma(1 + o(1))$. Substituting $t = u + \gamma s$ in (5.7) therefore gives $\pi^{-1} \int_{-K}^{K} (1+s^2)^{-1}\,ds + o(1) = p_K + o(1)$. The two events have intersection probability at least

$$q_K + o(1), \qquad q_K = 2p_K - 1 > 0. \tag{6.3}$$

Choose R so that the Poisson mass of $|y - x| > Rh$ is less than $q_K/8$, and let

$$E_n = \{x \in E : (p_h * \mathbf{1}_{\mathbb{R}\setminus E})(x) < q_K/8\}.$$

The approximate identity theorem gives $|E \setminus E_n| \to 0$. For all sufficiently large n, (6.3) is greater than $q_K/2$. For $x \in E_n$, the two excluded sets $\mathbb{R} \setminus E$ and $\{|y - x| > Rh\}$ have total Poisson mass less than $q_K/4$. Hence there exists $y \in E$ with $|y - x| \le Rh$ and both events true. We may take y away from the nodes. Since $m(y) \pm g(y)$ both lie in an interval of length $2K\gamma_h(x)$,

$$|g(y)| \le K\gamma_h(x).$$

Equations (5.3) and (6.1) give

$$H^\eta(y) \le H(y) = \lambda_n(y)|g(y)| \le cK(\log n)\gamma_h(x).$$

Together with (6.2), this yields

$$H^\eta(x) \le (1 + Rh/\eta)^2 H^\eta(y) \le cK(1 + Rh/\eta)^2(\log n)\gamma_h(x) \qquad (x \in E_n). \tag{6.4}$$

Let T_n be convolution with the symmetric probability kernel

$$k_n(t) = \frac{1}{b \log n} \int_\eta^1 p_{s-h}(t)\,\frac{ds}{s}.$$

Its integral is one, since $\int_\eta^1 ds/s = b \log n$. Equation (5.4) then shows that

$$H^\eta = \frac{2b}{\pi}(\log n) T_n \gamma_h.$$

Thus (6.4) becomes

$$\frac{T_n\gamma_h(x)}{\gamma_h(x)} \le \frac{1}{A_n} \quad (x\in E_n), \qquad A_n = \frac{2b}{\pi c K}(1+Rh/\eta)^{-2} \longrightarrow A>1. \tag{6.5}$$

Integrate (6.5) over E_n, discard the contribution from $y\notin E_n$, and symmetrize. Positivity of γ_h, symmetry of k_n, and $(r+r^{-1})/2\ge 1$ give

$$\frac{|E_n|}{A_n} \ge \iint_{E_n\times E_n} k_n(x-y)\frac{\gamma_h(y)}{\gamma_h(x)}\,dy\,dx \ge \iint_{E_n\times E_n} k_n(x-y)\,dy\,dx. \tag{6.6}$$

The kernels k_n form an approximate identity. In fact, for every fixed $\delta>0$,

$$\int_{|t|>\delta} k_n(t)\,dt \le \frac{C}{b\log n}\int_\eta^1 \min(1,s/\delta)\,\frac{ds}{s} \longrightarrow 0.$$

It follows that

$$\iint_{E\times E} k_n(x-y)\,dy\,dx \longrightarrow |E|.$$

Replacing E by E_n changes this integral by at most $2|E\setminus E_n|$. Taking limits in (6.6) therefore yields

$$\frac{|E|}{A}\ge |E|,$$

contrary to $|E|>0$ and $A>1$. This proves the assertion. □

Corollary. *Let $E\subset(-1,1)$ be measurable with $|E|>0$, and let $0<c<2/\pi$. For all sufficiently large n, there is $x\in E$ such that $\lambda_n(x)>c\log n$.*

Proof. Otherwise $\lambda_n(y)\le c\log n$ for every $y\in E$ and every n in an infinite set S. The argument above applies to these rows, with all limits taken as $n\to\infty$ through S, and gives the same contradiction $|E|/A\ge|E|$ with $A>1$. □

7 Failure of uniform additive constants

We prove Theorem 2. Smooth on $I=[-1,1]$ will mean extendible to a smooth function on a neighborhood of I.

7.1 The compact set

For $I=[-1,1]$, write

$$Lf(x) = \int_I \frac{f(x)-f(y)}{|x-y|}\,dy.$$

In particular,

$$\|Lf\|_\infty \le 2\|f'\|_\infty \qquad (f\in C^1(I)). \tag{7.1}$$

Fix $A>1$, and put

$$\rho = e^{-1024(A+40)}, \qquad \delta_k=\rho\,2^{-k}, \qquad m_k=\left\lceil e^{1/\delta_k}\right\rceil.$$

Starting with I, replace every retained interval of length ℓ_{k-1} at stage k by m_k equally long closed intervals, retaining both endpoints of the parent. The children and intervening open gaps have respective lengths

$$\ell_k = \frac{(1-\delta_k)\ell_{k-1}}{m_k}, \qquad b_k = \frac{\delta_k \ell_{k-1}}{m_k - 1}, \qquad \ell_0 = 2.$$

Let F be the intersection of the retained sets and $U = I \setminus F$. Thus F is compact, contains both endpoints, and has empty interior, since $\ell_k \to 0$. The set U is a dense open union of disjoint gaps, and

$$|U| \le 2 \sum_k \delta_k = 2\rho. \tag{7.2}$$

We need the following two properties:

$$|F \cap (x-r, x+r)| \ge \kappa r \quad (x \in F,\ 0 < r \le 1), \qquad \kappa = 1/32, \tag{7.3}$$

and

$$\int_U \frac{dy}{|x-y|} = \infty \qquad (x \in F). \tag{7.4}$$

For (7.3), choose the first k for which $\ell_k \le r/16$, and let J be the stage $k-1$ parent containing x, with $|J| > r/16$. At least half the length of any retained interval belongs to F, since every subsequent product of retained proportions is at least $1-\rho > 1/2$. If $|J| \le r$, then $J \subset [x-r, x+r]$, proving (7.3). Otherwise $J \cap [x-r, x+r]$ has length $s \ge r$. Write $a = \ell_k$, $b = b_k$, $\delta = \delta_k$. Then

$$b \le a, \qquad b/(a+b) \le 2\delta.$$

The gaps intersecting an interval of length s have total length at most $2\delta s + 2b$. The two partially intersected children have total length at most $2a$. The full children inside this interval therefore have total length at least

$$(1-2\delta)s - 2(a+b) \ge r/2.$$

Their intersections with F have total length at least $r/4$, as required.

For (7.4), in the stage $k-1$ parent containing x, one side of its child contains at least $(m_k-1)/2$ gaps. Use the first $q = \lfloor m_k/8 \rfloor$ on that side. The far endpoint of the j-th such gap is at distance at most $4j\ell_{k-1}/m_k$ from x, and its length is at least $\delta_k \ell_{k-1}/m_k$. These gaps contribute at least

$$\frac{\delta_k}{4} \sum_{j=1}^{q} \frac{1}{j} \ge \frac{\delta_k}{4}(\log m_k - \log 16) \ge \frac{1}{8}.$$

The gaps used at different stages are disjoint. Summation proves (7.4).

7.2 An explicit logarithmic integral estimate

On each gap (b, c), define

$$r(x) = \frac{(x-b)(c-x)}{c-b},$$

and put $r = 0$ on F. If $D(x) = \mathrm{dist}(x, F)$, then

$$r(x) \le D(x) \le 2r(x) \qquad (x \in U). \tag{7.5}$$

The function r is globally 1-Lipschitz: its derivative has absolute value at most one on each gap, and points in distinct gaps can be compared through a point of F between them.

Put

$$B = e^8, \qquad \alpha = 1/256, \qquad p(t) = [\log(B/t)]^{-\alpha}, \qquad u(x) = p(r(x)) \ (x \in U), \quad u = 0 \ (x \in F). \tag{7.6}$$

Then u is continuous on I, bounded, positive and smooth in each gap, and vanishes on F. In particular, $Lu(x)$ is an absolutely convergent integral for every $x \in U$.

We prove the quantitative bound

$$\frac{Lu(x)}{u(x)} \geq \frac{1}{512} \log \frac{1}{4r(x)} - 32 \qquad \text{if } \log(B/r(x)) \geq 16. \tag{7.7}$$

All estimates below apply uniformly up to the endpoints of I.

Fix such an x, and write $s = r(x)$, $T = \log(B/s)$, $a_0 = 4s$. For $t \in [0, 2]$, let

$$M_I(t) = |I \cap (x - t, x + t)|, \qquad M_U(t) = |U \cap (x - t, x + t)|.$$

Choose $z \in F$ with $|x - z| = D(x) \leq 2s$. If $4s \leq t \leq 2$, the ball of radius $t/2$ about z, intersected with I, is contained in the ball of radius t about x. By (7.3), the latter contains at least $\kappa t/2$ of F. Since $M_I(t) \leq 2t$,

$$M_U(t) \leq (1 - \eta) M_I(t), \qquad \eta = \kappa/4 = 1/128 \quad (4s \leq t \leq 2). \tag{7.8}$$

First consider $|y - x| < 4s$. For $|y - x| < s/2$, Lipschitz continuity of r gives $r(y) \in [s/2, 3s/2]$. On this range $p'(r(y)) \leq 4p(s)/s$, by the derivative formula for p and $T \geq 16$. The integral of the absolute difference quotient over this part is thus at most $4p(s)$. On $s/2 \leq |y - x| < 4s$, we have $r(y) \leq 5s$ and

$$p(5s)/p(s) = \left(\frac{T}{T - \log 5}\right)^{\alpha} \leq 2.$$

The absolute difference is at most $3p(s)$, so this part contributes at most $6 \log 8 \, p(s) < 18p(s)$. In particular, the whole near contribution to $Lu(x)$ is at least $-24p(s)$.

For the far contribution, set $l_- = 1 + x$, $l_+ = 1 - x$, and

$$H = \sum_{\nu \in \{-,+\}} \max\{\log(l_\nu/(4s)), 0\}.$$

The positive part is exactly $p(s)H$. For the negative part, $r(y) \leq s + |x - y| \leq (5/4)|x - y|$, so it is bounded by

$$\int_{a_0}^{2} f(t) \, dM_U(t), \qquad f(t) = \frac{p(5t/4)}{t}.$$

The function f is decreasing. Stieltjes integration by parts, dropping the nonpositive lower-end boundary term and using (7.8), gives

$$\int_{a_0}^{2} f(t) \, dM_U(t) \leq (1 - \eta) \left[f(a_0) M_I(a_0) + \int_{a_0}^{2} f(t) \, dM_I(t) \right]. \tag{7.9}$$

The first term on the right, including its factor, is at most $4p(s)$.

Since $dM_I(t) = 1_{\{t < l_-\}} dt + 1_{\{t < l_+\}} dt$, the remaining term splits into at most two integrals. For a side with $l_\nu > 4s$, put

$$h = \log(l_\nu/(4s)), \qquad T_0 = T - \log 5.$$

Then $0 \le h < T_0$, since $T_0 - h = \log(4B/(5l_\nu)) > 0$, and

$$\int_{4s}^{l_\nu} \frac{p(5t/4)}{t}\,dt = \int_0^h (T_0 - z)^{-\alpha}\,dz \le \frac{T_0^{-\alpha}}{1-\alpha}\,h. \tag{7.10}$$

The last inequality follows from $(1-z)^{1-\alpha} \ge 1 - z$ for $0 \le z \le 1$.

Finally, concavity of the power α, $\log 5 < 2$, and $T \ge 16$ give

$$\begin{aligned} \frac{1-\eta}{1-\alpha}\frac{T_0^{-\alpha}}{p(s)} &= \frac{254}{255}\left(\frac{T}{T-\log 5}\right)^{1/256} \\ &\le \frac{254}{255}(8/7)^{1/256} \le \frac{254}{255}\left(1 + \frac{1}{1792}\right) \le 1 - \frac{1}{512}. \end{aligned} \tag{7.11}$$

Thus the negative far contribution is at most $4p(s) + (1 - 1/512)p(s)H$. Including the near part gives $Lu(x)/u(x) \ge H/512 - 28$. At least one of l_-, l_+ is at least one, so $H \ge \log(1/(4s))$. This proves (7.7), with room in its constant.

Every gap has length at most $|U| \le 2\rho$, so $r(x) \le \rho/2$. Our choice of ρ ensures $T \ge 16$, and (7.7) yields

$$\frac{Lu(x)}{u(x)} \ge 2(A+40) - \frac{\log 2}{512} - 32 > A + 2 \qquad (x \in U). \tag{7.12}$$

It also shows that $Lu(x)/u(x) \to +\infty$ as x approaches either endpoint of any fixed gap.

7.3 Smooth positive functions on the whole interval

Choose a nondecreasing smooth $q_0 : \mathbb{R} \to [0,1]$, equal to zero on $(-\infty, 1]$ and one on $[2, \infty)$. On a gap (b, c), set

$$\chi_\tau(x) = q_0((x-b)/\tau)q_0((c-x)/\tau),$$

and put $\chi_\tau = 0$ on F. Only finitely many gaps, those of length greater than 2τ, contribute. Therefore χ_τ is smooth on I; it vanishes for $D \le \tau$, equals one for $D \ge 2\tau$, and increases pointwise to 1_U as $\tau \downarrow 0$.

For $x \in F$, define

$$A_\tau(x) = \int_U \frac{\chi_\tau(y)}{|x-y|}\,dy.$$

There is a direct uniform bound

$$A_{b_N/8}(x) \ge N/16 \qquad (x \in F). \tag{7.13}$$

Indeed, b_k decreases strictly with k. If $k \le N$, the middle half of every stage k gap has distance at least $b_k/4 \ge 2(b_N/8)$ from its endpoints, so $\chi_{b_N/8} = 1$ there. The argument proving (7.4), restricted to these middle halves, gives at least $1/16$ from each of the first N stages.

Let $\epsilon_j = 1/j$, $j \ge 2$, and put

$$d_j = \exp(8 - (2j)^{256}), \qquad C_j = 2(2j)^{256}.$$

Since $r \le D$ and $p(d_j) = 1/(2j)$, one has $u(y) \le \epsilon_j/2$ when $D(y) \le d_j$. Using this for $|x-y| < d_j$, and $u \le 1$ on the remainder, gives, uniformly for $x \in F$,

$$\int_U \frac{\chi_\tau(y)u(y)}{|x-y|}\,dy \le \frac{\epsilon_j}{2}A_\tau(x) + C_j. \tag{7.14}$$

Here the far integral is at most $2\log(2/d_j) \le C_j$.

Now specify the increasing integers and decreasing cutoffs

$$k_j = \lceil 32(A+1) \rceil + 64j(2j)^{256} + j, \qquad \tau_j = b_{k_j}/8.$$

By (7.13),

$$\frac{\epsilon_j}{2} A_{\tau_j}(x) - C_j \ge \frac{k_j}{32j} - C_j \ge (A+1)\epsilon_j \qquad (x \in F).$$

Define

$$v_j = \chi_{\tau_j} u + (1 - \chi_{\tau_j})\epsilon_j. \tag{7.15}$$

This function is strictly positive and smooth on a neighborhood of I: only finitely many compact portions of gaps contribute, u is smooth and positive there, and the function equals ϵ_j near their boundaries and near the endpoints of I.

At $x \in F$, (7.14) gives $Lv_j(x) \ge (A+1)v_j(x)$. At every fixed $x \in U$, eventually $v_j = u$ on a neighborhood of x. Also $v_j \to u$ in $L^1(I)$, by bounded convergence. The integrals outside that neighborhood then converge, so

$$\frac{Lv_j(x)}{v_j(x)} \longrightarrow \frac{Lu(x)}{u(x)} > A + 2 \qquad (x \in U). \tag{7.16}$$

We have therefore constructed positive smooth functions such that

$$\frac{Lv_j(x)}{v_j(x)} \ge A + 1 \quad \text{eventually for every fixed } x \in I. \tag{7.17}$$

The eventual index is allowed to depend on x.

7.4 From amplitudes to interpolation nodes

We prove an upper estimate with an absolute additive constant.

Lemma 9. *Let $h(z) = \sum_{r=0}^{d} h_r z^r$ have real coefficients, no zero on the closed unit disk, and $h(0) > 0$. Set*

$$v(\cos\theta) = |h(e^{i\theta})|^{-1}.$$

In this subsection, T_k denotes the Chebyshev polynomial of the first kind, defined by $T_k(\cos\theta) = \cos(k\theta)$. For $n > d$, define the real polynomial

$$P_n(x) = \sum_{r=0}^{d} h_r T_{n-r}(x). \tag{7.18}$$

For all sufficiently large n, this polynomial has exactly n simple roots in $(-1, 1)$. The corresponding Lebesgue function satisfies, on any fixed $K \Subset (-1, 1)$,

$$\lambda_n(x) \le \frac{2}{\pi} \log n + 3 - \frac{Lv(x)}{\pi v(x)} \qquad (x \in K) \tag{7.19}$$

for all sufficiently large n, with the threshold allowed to depend on h, K.

Proof. The analytic logarithm of h, chosen real at zero, is real on the real diameter. Its imaginary part $\psi(\theta) = \arg h(e^{i\theta})$ thus satisfies $\psi(0) = \psi(\pi) = 0$. We have the exact identity

$$P_n(\cos\theta) = |h(e^{i\theta})| \cos(n\theta - \psi(\theta)).$$

For $n > \|\psi'\|_\infty$, the phase increases strictly from 0 to $n\pi$. The polynomial has degree exactly n, since its leading term comes from $h_0 T_n$, with $h_0 > 0$. This gives the asserted n simple roots. At their angles θ_k,

$$|P_n'(\cos\theta_k)| = |h(e^{i\theta_k})| \frac{n - \psi'(\theta_k)}{\sin\theta_k}. \tag{7.20}$$

Put $s = (n\theta - \psi(\theta))/n$, let $\theta = \Theta_n(s)$ be the inverse, and set $X_n(s) = \cos\Theta_n(s)$. The roots correspond to $s_k = (k - \frac{1}{2})\pi/n$. Moreover $X_n \to \cos$ in $C^3([0,\pi])$. At a point $x = X_n(s)$ other than a node, put $b = |\cos(ns)|$. Formula (7.20) gives

$$\lambda_n(x) = \frac{b}{nv(x)} \sum_{k=1}^{n} \frac{v(X_n(s_k))|X_n'(s_k)|}{|X_n(s) - X_n(s_k)|}. \tag{7.21}$$

Subtract the simple singularity by defining

$$R_n(s,t) = \frac{v(X_n(t))|X_n'(t)|}{|X_n(s) - X_n(t)|} - \frac{v(X_n(s))}{|s-t|}.$$

For s in a fixed compact subinterval of $(0,\pi)$, this function of t has uniformly bounded variation, with at most a bounded jump at $t = s$. To verify this near s, factor $X_n(t) - X_n(s) = (t-s)D_n(s,t)$. Both D_n and X_n' are uniformly bounded away from zero there; the numerator remaining after subtraction vanishes at $t = s$, so division leaves a uniformly C^1 function on each side, multiplied by $\operatorname{sgn}(t-s)$. Away from s, the denominator stays uniformly away from zero. Midpoint quadrature therefore has error $O(1/n)$, uniformly also when s is arbitrarily close to a quadrature point.

Changing variables $y = X_n(t)$ in the integral, and first deleting $(s-\epsilon, s+\epsilon)$, gives

$$\begin{aligned} \int_0^\pi R_n(s,t)\,dt &= -Lv(x) + v(x)\log\frac{1-x^2}{X_n'(s)^2 s(\pi - s)} \\ &= -Lv(x) + v(x)\left[2\log(1 - \psi'(\theta)/n) - \log(s(\pi - s))\right]. \end{aligned} \tag{7.22}$$

The cancellation follows because the deleted distances in the y variable are $|X_n'(s)|\epsilon + O(\epsilon^2)$ on both sides.

Write

$$\tau = \frac{ns}{\pi} + \frac{1}{2} = m + t, \qquad m = \lfloor\tau\rfloor, \quad 0 < t < 1.$$

Since $(s - s_k)n/\pi = \tau - k$, midpoint quadrature in (7.21) gives explicitly

$$\lambda_n(x) = \frac{b}{\pi}\left[\sum_{k=1}^{n} \frac{1}{|\tau - k|} + \frac{1}{v(x)}\int_0^\pi R_n(s,t')\,dt'\right] + O(1/n).$$

On the compact interval under consideration, $1 \le m \le n-1$ for large n, and $b = \sin(\pi t)$. Splitting the elementary harmonic sum at m gives

$$\sum_{k=1}^{n} \frac{1}{|\tau - k|} \le \log(m(n-m)) + 2 + \frac{1}{t} + \frac{1}{1-t}.$$

Also, uniformly in this compact interval,

$$\log(m(n-m)) - \log(s(\pi - s)) = 2\log n - 2\log\pi + O(1/n).$$

Substituting these estimates and (7.22) into (7.21), and using $b/t \le \pi$, $b/(1-t) \le \pi$, yields

$$\lambda_n(x) \le b\left[\frac{2}{\pi}\log n - \frac{Lv(x)}{\pi v(x)} + \frac{2-2\log\pi}{\pi}\right] + 2 + O(1/n).$$

For large n the bracket is nonnegative, uniformly on K. We may replace $b \le 1$ by one. Since $2 + (2-2\log\pi)/\pi < 3$, this proves (7.19) away from the nodes. At the nodes, $\lambda_n = 1$, and (7.19) follows for large n as well. $\square$

Every positive smooth function on I can be approximated in $C^1(I)$ by amplitudes occurring in this lemma. Indeed, approximate its inverse square by a strictly positive real polynomial $p(x)$. The Laurent polynomial $p((z+z^{-1})/2)$ has reciprocal and conjugate pairs of zeros, with none on the unit circle. Choosing the zeros outside the circle gives a real polynomial h, zero-free on the closed disk, such that

$$p(\cos\theta) = |h(e^{i\theta})|^2.$$

The scalar factor is positive because both sides are positive on the circle; choose $h(0) > 0$. Thus $p^{-1/2}$ is the required amplitude. Polynomial approximation in C^1 follows by uniformly approximating the derivative and then integrating. For a fixed strictly positive v, (7.1) also shows that sufficiently accurate C^1 approximation gives arbitrarily accurate uniform approximation of Lv/v.

7.5 Assembly of the array

Proof of Theorem 2. Fix $M > 0$, and carry out Subsections 7.1–7.3 with $A = \pi(M+4)$. For each $j \ge 2$, choose an amplitude $\widetilde{v}_j = |h_j|^{-1}$ as in Subsection 7.4 so close to v_j that

$$\left\|\frac{L\widetilde{v}_j}{\widetilde{v}_j} - \frac{Lv_j}{v_j}\right\|_\infty \le 1/j. \tag{7.23}$$

Let $K_j = [-1+1/j, 1-1/j]$. Choose a strictly increasing sequence of integers N_j, each larger than $\deg h_j$ and the threshold in Lemma 9 for h_j, K_j. For $N_j \le n < N_{j+1}$, use as row n the n roots of (7.18) with $h = h_j$. Fill the finitely many earlier rows with Chebyshev roots.

Fix $x \in (-1,1)$. The block index j tends to infinity with n, and $x \in K_j$ for all large j. Equations (7.17) and (7.23) give $L\widetilde{v}_j(x)/\widetilde{v}_j(x) \ge A + 1 - 1/j$ eventually. Hence (7.19) gives, for all sufficiently large n,

$$\lambda_n(x) \le \frac{2}{\pi}\log n + 3 - \frac{A+1-1/j}{\pi} < \frac{2}{\pi}\log n - M.$$

This proves (1.3) and the failure of an absolute additive constant.

For the final assertion, (7.16), (7.19), and (7.23) give at every fixed $x \in U$

$$\limsup_{n\to\infty}\left(\lambda_n(x) - \frac{2}{\pi}\log n\right) \le 3 - \frac{Lu(x)}{\pi u(x)}. \tag{7.24}$$

Choose a gap (b,c) with both endpoints in $(-1,1)$. By (7.7), $Lu(x)/u(x) \to +\infty$ as $x \downarrow b$. Given any finite C, there is a nonempty interval $(b, b+\sigma)$ on which the right side of (7.24) is strictly less than $-C-1$. This interval is disjoint from G_C, proving the second assertion. $\square$

Use of generative AI

GPT-6 Astra was used to generate proofs and draft the manuscript; GPT-5.6 Sol and Claude Opus 5 were used for editorial review. An independent Codex agent also reviewed the complete proof of Theorem 2. Theorem 1(ii), including the arguments of Sections 5–6, and the set corollary have been formalized in Lean 4 with Mathlib using OpenAI Codex (GPT-6). The corollary is `Erdos1132.eventually_exists_lower_bound`, derived from `uniform_low_set_null` with an upper bound assumed for infinitely many rows. The source and build instructions are at `https://github.com/FireflySentinel/erdos-1132`. Selected lemmas of Section 7, including the conditional assembly step, are also formalized; their precise scope and hypotheses are documented with the source. Theorem 1(i) and the complete Theorem 2 remain outside the formalization.

References

[1] S. Bernstein, Sur la limitation des valeurs d'un polynôme $P_n(x)$ de degré n sur tout un segment par ses valeurs en $(n+1)$ points du segment, *Izv. Akad. Nauk SSSR*, Ser. VII (1931), no. 8, 1025–1050.

[2] T. F. Bloom, *Erdős Problem #1132*, `https://www.erdosproblems.com/1132`. Accessed 5 September 2026.

[3] L. Brutman, Lebesgue functions for polynomial interpolation—a survey, *Ann. Numer. Math.* **4** (1997), 111–127.

[4] P. Erdős, Problems and results on the convergence and divergence properties of the Lagrange interpolation polynomials and some extremal problems, *Mathematica (Cluj)* **10** (1968), 65–73. Original paper.

[5] P. Erdős, Problems and results on the theory of interpolation. II, *Acta Math. Acad. Sci. Hungar.* **12** (1961), 235–244.

[6] P. Erdős and P. Turán, An extremal problem in the theory of interpolation, *Acta Math. Acad. Sci. Hungar.* **12** (1961), 221–234.

[7] P. Erdős and P. Vértesi, On the almost everywhere divergence of Lagrange interpolatory polynomials for arbitrary system of nodes, *Acta Math. Acad. Sci. Hungar.* **36** (1980), 71–89.

[8] G. Faber, Über die interpolatorische Darstellung stetiger Funktionen, *Jahresber. Deutsch. Math.-Verein.* **23** (1914), 192–210.

[9] T. Tao, *Local Bernstein theory, and lower bounds for Lebesgue constants*, arXiv:2603.21453v3 (2026).

[10] P. Vértesi, Optimal Lebesgue constant for Lagrange interpolation, *SIAM J. Numer. Anal.* **27** (1990), 1322–1331. doi:10.1137/0727075.